

# Measurement of 3D displacement fields from few tomographic projections

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## ABSTRACT

The present paper aims at providing 3D volume images of a *deformed* specimen based on i) a full 3D image describing the reference state as obtained *e.g.*, from conventional computed tomography and ii) the 3D displacement field accounting for its motion. The displacement field, which is described by much fewer degrees of freedom than the specimen volume itself, is here proposed to be determined from very few projections. The reduction in number of needed projections may be larger than two orders of magnitude. In the proposed approach, the displacement field is described over an unstructured mesh composed of tetrahedra with linear shape functions. The mesh is based on the reconstructed reference volume so that it provides a faithful and accurate description of the specimen, including its boundary. Nodal displacements are determined from the minimization of the quadratic difference between the computed projections of the deformed configuration and the acquired projections (radiographs) for the selected orientations. Well-posedness of the problem requires the number of kinematic unknowns to be small. However, in cases where the geometry is complex, the displacement field may call for many parameters. To deal with such conflicting demands it is proposed to use a regularization based on the mechanical modeling of the displacement field using a linear elastic description.

**Keywords:** Tomographic reconstruction, Non-rigid registration, Digital volume correlation, Elastic regularization

## 1. INTRODUCTION

Akin to video compression<sup>1</sup> exploiting temporal persistence allows for big savings in information storage as only “differences” are to be recorded to retrieve successive frames. Differences being small and sparse, they can be very efficiently compressed. Our aim is the description of the deformation of a mechanically loaded sample within a tomograph. It is proposed to use a parameterization of the time evolution of a 3D tomographic image through its motion, to obtain a compressed, though not lossy, description. Moreover, as the description is reduced to a few parameters, the reconstruction itself may be reconsidered. The determination of the kinematic parameters requires much less projections (*i.e.*, radiographs) than those needed to produce a standard 3D image, provided the reconstruction has been performed once to make the reference volume available. Furthermore, using a mechanical modeling, the motion itself can be condensed into an even smaller set of parameters, even for a complex geometry or loading. Thus, at the expense of having to read from the radiographic projections the expression of the mechanical description on the reference volume, a very drastic reduction of the number of projections can be achieved. This reduction can be loss-less provided the mechanical model itself is reliable. Hence, such approaches open a new pathway to efficient 4D tomography, where non-rigid motions are key rather than obstacle in contrast to motion correction or stroboscopic approaches developed recently.<sup>2</sup>

The importance of tomography for material science has been reviewed by Maire and Withers,<sup>3</sup> emphasizing the quantitative character of 3D imaging as being a very significant evolution in the recent years. In particular the need for addressing fast morphological changes has led to impressive developments. Reduction in tomographic acquisition time has led to the possibility of tracking motion at a fast rate through ultrafast tomography.<sup>4</sup> For instance, ductile fracture has been fully imaged at a rate of 20 Hz.<sup>5</sup> This instrumentation progress is quite

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remarkable but is still based on classical tomographic reconstruction with a full set of projections. Thus, the combination of these spectacular acquisition progress with the proposed approach to reduce the number of projections should allow for even greater temporal resolution.

Let us also note that measuring 3D displacement fields in solids from tomographic images is a mature technique, derived with very modest changes from 2D non-rigid image registration, well-known as “Digital Image Correlation” (DIC) in solid mechanics.<sup>6</sup> In 3D, this “Digital Volume Correlation” (DVC) technique has been initiated by Bay *et al.*<sup>7</sup> It requires the full reconstruction of the reference and deformed volumes to measure the displacement field. It consists in determining the displacement field that allows the deformed image corrected for its motion to be matched with the reference image. Key to this registration operation is the fact that the kinematics can be described by a smoothly varying displacement field, with a good separation of scales between the morphological patterns of the microstructure and the characteristic scales of displacements. Thus, the motion description is usually parameterized with a spatially *coarse* description.

The *local* approach to DVC is based on a partition of the image into sub-volumes, whose kinematics is often described as a displacement field based on a low order polynomial of its coordinate, and all sub-volume center displacements form a discretization of the displacement field.<sup>8</sup> The *global* approach proposes a smoother description (*e.g.*, typically continuous displacement field) that spans over the entire region of interest and where the interdependence of the different kinematic degrees of freedom calls for a global determination of their values.<sup>9</sup> In spite of these differences, both approaches lead to a coarse grid or mesh that accounts for the motion in space. In quantitative terms, coarseness may refer to the fact that the displacement field is decomposed over a regular cubic mesh whose individual element may consist of  $10^3$ -voxels, to mention a typical order of magnitude, still with an uncertainty much smaller than 1 voxel (standard values may be in the range of 0.1 voxel). Accounting for the vector nature of displacements, the discretization of the displacement field calls for no more than 0.3% of the information content of the 3D volume itself (described as 1 gray-level per voxel). This very small ratio provides a very high redundancy of DVC, allowing partly for uncertainty reduction, and providing a possible validation of the measurement. Expressed in those terms, the present proposal is to invest part of this redundancy to save on the number of projections by including the reconstruction step in the analysis.

Section 2 presents an experimental case, namely a cast iron specimen in which a fatigue crack is propagating, which will be used all along this paper to illustrate the proposed methodology. Projection-based Digital Volume Correlation, (P-DVC), which is proposed hereinto extract kinematic information from few projections, is introduced in Section 3 in conjunction with tomographic reconstruction<sup>10</sup> and global DVC algorithm.<sup>9</sup> Elastic regularization, discussed in Section 4, provides an easily tunable way of reducing the number of kinematic unknowns to be determined, and makes the displacement discretization in a finite-element mesh more transparent to the analysis. Finally, Section 5 proposes a discussion of the results and explores the robustness in terms of mesh sensitivity, choice of projection, convergence performance and uncertainty.

## 2. EXPERIMENTAL TEST CASE

To illustrate the proposed methodology, an experimental test case will be used in the following sections. This test case is based on a study of fatigue crack propagation in a cast-iron specimen. Fatigue consists in cyclic loading a sample containing a crack, where at each cycle a very confined zone about the crack front undergoes plastic deformations, and the accumulation of these plastic strains leads to a very slow crack growth (at each cycle the crack progresses by a very small amount). The material contains nodular graphite inclusions, well dispersed in the iron matrix, that are clearly resolved in the reference image scanned at the European Synchrotron Radiation Facility (ID 19 beamline). 600 projections are needed to reconstruct the specimen volume at a resolution of 5.4  $\mu\text{m}$ . The sample size is  $1.79 \times 1.84 \text{ mm}^2$  in cross section and 2.77 mm in height. It is mounted in a tensile machine and the reference image is acquired under a very small load (12 N) applied to limit the effect of possible plays in the testing set up. The deformed state is obtained when the specimen is subjected *in situ* to a tensile force of 140 N. From the reference image, shown in Figure 1a, the crack surface can be segmented (Figure 1b).

In this particular case, the kinematics was measured using global DVC and hence enough (*i.e.*, 600) projections have been acquired in the deformed state so that a full classical reconstruction could be performed. This will serve as a gauge to evaluate the performance of the proposed P-DVC approach, where only 2 out of the 600 projections will be used (unless otherwise stated).

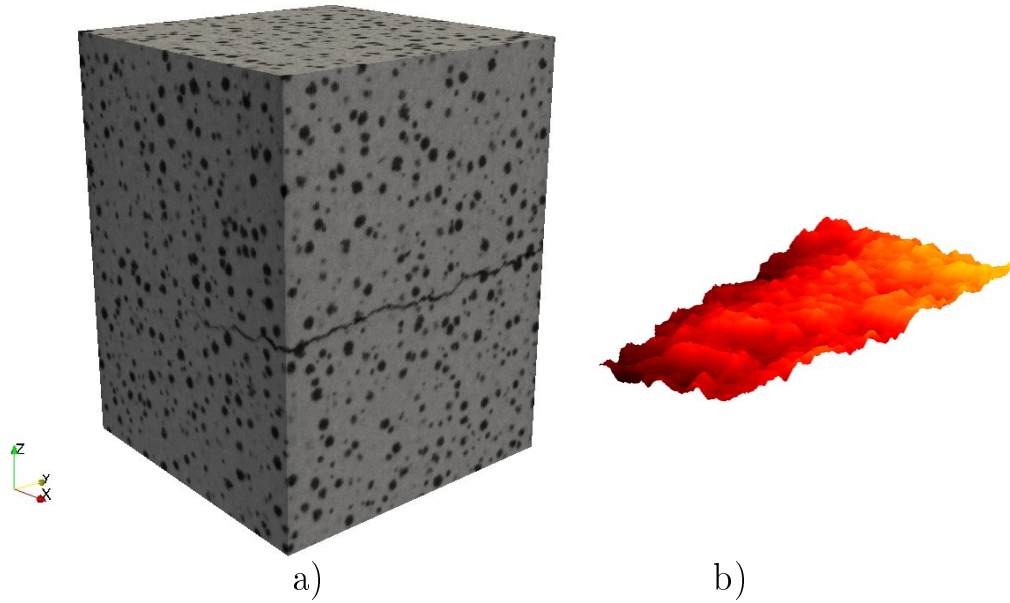


Figure 1. (a) Reference image of the cast iron specimen. The tensile axis (vertical in the figure) loads the crack in mode I (*i.e.*, opening). (b) Segmentation of the crack surface. The color indicates the  $z$ -coordinate along the vertical direction

### 3. PROPOSED APPROACH

#### 3.1 Tomographic reconstruction

There is a vast literature on techniques to relate the acquired radiographs to the reconstructed object. The reader is referred for instance to Kak and Slaney's book<sup>10</sup> for a global overview of the topic. We will simply mention here some elementary considerations to specify our notations.

The sample is mounted on a turn table, whose rotation axis  $z$  is perpendicular to the X-ray beam. It allows the sample to be exposed to the beam for a range of angles,  $0 \leq \phi \leq \pi$ , at equally spaced intervals, here 600, for which a radiograph  $I(r, z, \phi)$  is acquired from a detector, where  $r$  is an axis of the detector plane perpendicular to  $z$ . A flat field  $I_0(r, z)$  has been recorded (and possibly interpolated to account for beam intensity variations or other artifacts). From these two images, projections are computed as  $P(r, z, \phi) = -\log(I(r, z, \phi)/I_0(r, z))$ . Based on Beer-Lambert law, the projection is interpreted as the line integral of the absorption coefficient  $f(x, y, z)$  in the medium for the ray in the beam that hits the detector at position  $(r, z)$  when the sample is rotated by  $\phi$ . Mathematically,  $f$  and  $P$  can be transformed into each other without loss of information, from the Radon (projection), denoted  $\mathfrak{P}$  and the inverse Radon,  $\mathfrak{R}$ , transforms so that  $\mathfrak{P}[f] = P$  and  $\mathfrak{R}[P] = f$ .

Different softwares are available for tomographic reconstruction. Some of the following results are obtained with the algebraic reconstruction implementation available from the ASTRA toolbox.<sup>11</sup>

#### 3.2 Digital Volume Correlation

Classically DVC consists in relating two images  $f(\mathbf{x})$  and  $g(\mathbf{x})$  (where  $\mathbf{x} = (x, y, z)$  denotes vectors as bold symbols) through a displacement field  $\mathbf{u}(\mathbf{x})$ , assuming the microstructure is simply advected by the displacement, or  $f(\mathbf{x}) = g(\mathbf{x} + \mathbf{u}(\mathbf{x}))$ . However, noise and imaging artifacts prevent perfect match to be reached and hence the displacement is estimated from the minimizer of the difference between the corrected deformed image and the reference one, which is called residual. Hence, introducing the objective functional

$$\mathcal{T}_{DVC}[\mathbf{w}] = \|\mathbf{f}(\mathbf{x}) - \mathbf{g}(\mathbf{x} + \mathbf{w}(\mathbf{x}))\|^2 \quad (1)$$

the measured displacement field  $\mathbf{u}(\mathbf{x})$  obeys

$$\mathbf{u} = \underset{\mathbf{w} \in \mathcal{U}}{\text{Argmin}} \mathcal{T}_{DVC}[\mathbf{w}] \quad (2)$$

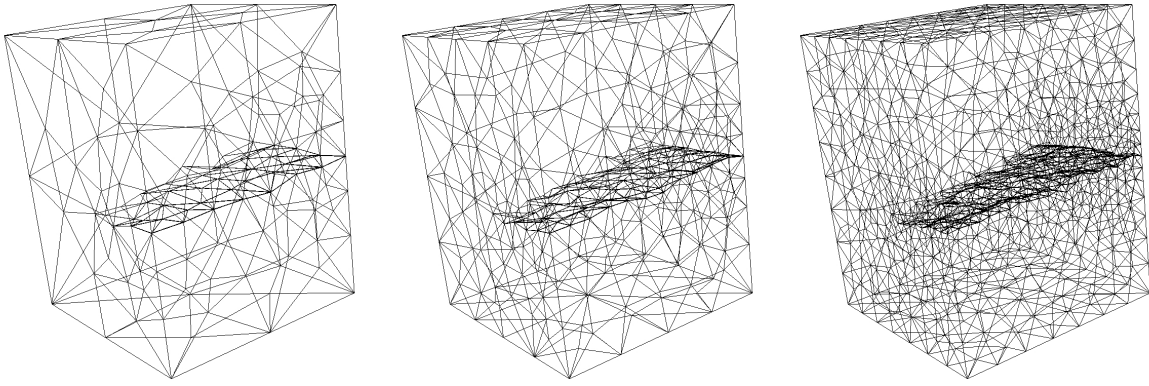


Figure 2. Three meshes: (a) coarse, (b) medium and (c) fine, which are fitted to the sample boundary including the crack and where a discontinuity is allowed. The crack geometry is derived from the segmentation shown in Figure 1(b)

where  $\mathcal{U}$  is a finite dimension space of displacement fields that relies on a specific discretization. A convenient form issued from mechanical modeling is to use a finite element framework to describe the kinematics. Hence, the medium  $f$  is meshed over a chosen region of interest. In 3D, it is common to use tetrahedral elements (4-noded tetrahedron where the displacement is linearly interpolated in the volume from the nodes). Thus the trial displacement field is discretized as  $\mathbf{w}(\mathbf{x}) = \sum_i \mathbf{W}_i N_i(\mathbf{x})$  where  $N_i(\mathbf{x})$  are the element shape functions valued 1 on node  $i$  and 0 on all other nodes, and  $\mathbf{W}_i$  is the vector displacement at node  $i$ . The collection of nodal displacement is gathered in a single column vector  $\{\mathbf{W}\}$ . In this case, the displacement fields in  $\mathcal{U}$  are piecewise linear and continuous fields. Sophisticated meshing strategies are readily available as this field is quite mature. The present work uses GMSH.<sup>12</sup>

In the following analysis, the reference volume is available. Thus the mesh can be tailored to follow tightly the boundaries including the crack. Crack opening is naturally allowed as the two opposite faces of the crack mouth are not connected. Figure 2 shows three meshes that conform to the actual geometry of the sample.

It is to be underlined that although this presentation emphasizes on the similarity between DVC and Finite-Element modeling as similar supports (*i.e.*, meshes) are used, major differences exist. Finite-element mechanical modeling aims at computing the best approximation to a system of partial differential equations in a finite dimensional space and, in the limit of a vanishing element size, the variational formulation allows the exact solution to be approached. Conversely, DVC exploits the image texture (*i.e.*, gray level distributions in space) in order to register the two images and when the element size becomes too small, the problem becomes ill-conditioned and then ill-posed. Hence, small uncertainties require large elements. Therefore a natural limit of DVC is the compromise between spatial resolution and small uncertainties.<sup>13</sup>

### 3.3 Projection-based DVC

P-DVC was introduced in Ref. 14. The same lines are followed herein with small variants. In particular, a Lagrangian description of the motion is preferred rather than the Eulerian as originally proposed.<sup>14</sup> The Lagrangian displacement field  $\mathbf{u}(\mathbf{x})$  is computed at the reference positions  $\mathbf{x}$ , and allows its correspondence to be computed in the deformed configuration  $\mathbf{X} = \mathbf{x} + \mathbf{u}(\mathbf{x})$ , whereas the Eulerian displacement field  $\mathbf{V}(\mathbf{X})$  would determine the displacement in the deformed configuration so that  $\mathbf{x} + \mathbf{V}(\mathbf{X}) = \mathbf{X}$ . Both descriptions are mathematically equivalent, but depending on the addressed problem, one may be more convenient than the other. The Lagrangian description is preferred because the starting mesh is finely adjusted to the geometry and structural features (such as the crack) of the reference state. However, it is necessary to deform the reference image to compute a full 3D image whose projection will be matched with the acquired one. This operation is performed in two steps. First, the mesh nodes  $\mathbf{x}_i$  are transported in the deformed space to positions  $\mathbf{X}_i = \mathbf{x}_i + \mathbf{u}(\mathbf{x}_i)$ . Then any voxel of the deformed space,  $\mathbf{X}$  is positioned within its element, based on  $\mathbf{X}_i$ , and from its element barycentric coordinates,  $\omega_i$  such that  $\mathbf{X} = \omega_i \mathbf{X}_i$  (using the isogeometric property of the element) the gray level

$\tilde{f}_{\mathbf{u}}(\mathbf{X}) = f(\omega_i \mathbf{x}_i)$  is computed. From this virtual 3D image a projection is readily accessible to compute  $\mathfrak{P}[\tilde{f}_{\mathbf{u}}]$  for any chosen angle  $\phi$ . Those computed projections can be compared with the actual projections of the deformed image  $P_g$ . P-DVC<sup>14</sup> is thus naturally formulated as finding the minimizer of the norm of differences between computed and measured projections, or, denoting

$$\mathcal{T}_{PDVC}[\mathbf{w}] = \|\mathfrak{P}[\tilde{f}_{\mathbf{w}}] - P_g\|^2 \quad (3)$$

the measured displacement field  $\mathbf{u}$  obeys

$$\mathbf{u} = \underset{\mathbf{w} \in \mathcal{U}}{\text{Argmin}} \mathcal{T}_{PDVC}[\mathbf{w}] \quad (4)$$

where the quadratic norm is the sum over a set of chosen angles  $\phi \in \Phi_{meas}$  of the sum over  $(r, z)$  clipped to a specific region of interest that spans over the entire sample projection in  $r$  and is a chosen interval in  $z$ . The corresponding region of interest in the deformed volume when transported in the reference one should be strictly contained in the meshed domain. In order to avoid having a partial information  $\tilde{f}$  along some rays, a masking procedure is introduced. It compares the projections in a region always smaller than the entire mesh projection in order to avoid the effect of phase contrast along the crack faces (due to steep absorption gradients).

One difficulty of this formulation is the fact that many kinematic degrees of freedom influence the very same area in a given projection. Disentangling these influences in order to solve for the displacement is achieved using a Newton-Raphson algorithm. However, akin to standard DVC, a fine mesh (*i.e.*, many degrees of freedom) implies a poor conditioning of the problem, hence large uncertainties and slow convergence. Additionally, the computation time increases fast with the number of mesh nodes as the construction of the deformed volume is to be carried out voxel-wise.

#### 4. ELASTIC REGULARIZATION

To circumvent the dilemma between the need for using few degrees of freedom, and yet to have a faithful description of the displacement field (displacement uncertainty is targeted to be much smaller than one voxel) a regularization scheme is introduced. However, this regularization should not be too restrictive or influential on legitimate degrees of freedom. Following earlier works in 2D<sup>15,16</sup> or 3D<sup>13,17</sup> a penalty for displacements not being the solution to a linear elastic problem over the same medium geometry is enforced. More precisely, an elastic problem is summarized as:

- i) a balance equation, which in the absence of body forces states that the stress field,  $\boldsymbol{\sigma}$ , is divergence-free, or  $\sigma_{ij,j} = 0$  for all  $i \in \{1, 2, 3\}$
- ii) a constitutive equation, Hooke's law, relating stress to strain,  $\boldsymbol{\epsilon}$ , through  $\sigma_{ij} = 2\mu\epsilon_{ij} + \lambda\epsilon_{kk}\delta_{ij}$ , where  $\lambda$  and  $\mu$  are two elastic constants
- iii) a compatibility condition, relating strains to displacements,  $\epsilon_{ij} = (1/2)(w_{i,j} + w_{j,i})$ .

Gathering those three relationships yields a second order differential equation obeyed by the displacement field

$$w_{i,jj} + \frac{(\mu + \lambda)}{\mu} w_{j,ji} = 0 \quad (5)$$

or in short  $\mathfrak{L}\mathbf{w} = 0$  where  $\mathfrak{L}$  is Lamé's operator. Within the finite element formalism, this same relationship takes the discrete form  $[\mathbf{B}][\mathbf{K}]\{\mathbf{W}\} = \{\mathbf{0}\}$ , where  $[\mathbf{K}]$  is the rigidity matrix,  $\{\mathbf{W}\}$  the vector of nodal displacements, and  $[\mathbf{B}]$  is a diagonal matrix valued 1 for internal or free-boundary nodes, and valued 0 for surface nodes where a displacement is prescribed (Dirichlet boundary). Elastic regularization consists in adding the penalty  $\mathcal{T}_{Reg}[\{\mathbf{W}\}] = \|[\mathbf{B}][\mathbf{K}]\{\mathbf{W}\}\|^2$  to the P-DVC functional  $\mathcal{T}_{PDVC}[\{\mathbf{W}\}]$ . These two functionals are to be normalized and weighted in a global objective functional

$$\mathcal{T}_{tot} = \mathcal{T}_{PDVC} + \eta \mathcal{T}_{Reg} \quad (6)$$

In order to make explicit the meaning of the weight  $\eta$ , it is instructive to consider a plane wave mode  $\hat{\mathbf{w}}(\mathbf{x}) = \mathbf{w}_0 \exp(i\mathbf{k}_0 \cdot \mathbf{x})$ . The scaling properties of these functionals are  $\mathcal{T}_{PDVC}[\hat{\mathbf{w}}] \propto |\mathbf{k}_0|^0$  and  $\mathcal{T}_{Reg}[\hat{\mathbf{w}}] \propto |\mathbf{k}_0|^4$ , so that  $\eta$  is written as

$$\eta = \xi^4 \frac{|\mathbf{k}_0|^4}{(2\pi)^4} \frac{\mathcal{T}_{PDVC}[\hat{\mathbf{w}}]}{\mathcal{T}_{Reg}[\hat{\mathbf{w}}]} \quad (7)$$

where  $\xi$  is a characteristic regularization length scale, which can be freely tuned. The regularization part plays the role of a low pass filter. High frequency modes of the displacement field with a wavelength less than  $\xi$  are dampened in order to get closer to the solution to a homogeneous elastic problem. In contrast, the low frequency modes are dictated by the projection matching. It is however to be noted that the kernel of  $\mathcal{T}_{Reg}$  is large. In particular, it is possible to prescribe any arbitrary displacement on the nodes  $i$  for which  $B_{ii} = 0$  (no summation), and still maintain a zero value for the regularization functional. Thus even when  $\xi \rightarrow \infty$ , that is when  $\mathbf{w}$  is exactly the solution to an elastic problem, the dimensionality of the space where  $\mathbf{w}$  is defined is large.

## 5. RESULTS AND DISCUSSION

The above procedure is applied to the test case that was described in Section 2. Because the mechanical behavior of the sample in the considered range of loading is essentially elastic, regularization is set to an infinite weight  $\xi \rightarrow \infty$ . Moreover, the boundary conditions on the two opposite top and bottom faces are set to rigid body motions in order to further simplify the problem. The difference in rigid body motion is sufficient to account for any global rigid body motion, and additionally to all loading modes that propagate to an infinite distance for a slender geometry. Thus in the following the six degrees of freedom of the rigid body motion of the top and bottom faces of the sample are set free, so that a total of twelve unknowns are to be determined. For each degree of freedom, the nodal displacements are computed from an elastic computation. The following results are obtained from only two projection directions. The probed region of interest in each projection is 400 voxels high, and spans laterally over the entire sample width. The best combination of parameters are first chosen to define our reference from which the sensitivity to different variants is discussed.

### 5.1 Assessment of baseline convergence

It is important to first assess the quality of the identified displacement field. In the general case, the only available information to validate convergence is the matching of the projections. The gauge with which those residuals are to be evaluated can be set by choosing as target projections those derived from the radiographs of the reference state. In this case, one should measure zero displacement, and hence the measured values give an order of magnitude of the uncertainty on the kinematic parameters. The measured root mean square RMS displacements (expressed in voxel size) are in this case 0.21, 0.18, 0.16 in the three directions.

Moreover, the projection residuals,  $\rho_0 = \mathfrak{P}[\tilde{f}_{\mathbf{w}}] - P_g$  are computed for the chosen projection directions. This test is not trivial since the reconstructed volume was computed from the 600 projections, and thus the selected angles for the P-DVC procedure represent a very small contribution. Noise and radiograph acquisition artifacts limit the accuracy of the reconstruction. In particular for the present test case, the treatment of flat fields providing the intensity of the radiographs in the absence of sample can be shown to play a key role in the reconstruction. The figure of merit for this application is related to the low norm of the residual field. However, it is noticed that the crack is clearly visible on the residual. This may be due to an imperfect segmentation, but it may also result from phase contrast artifacts that are not included in the reconstruction or projection operators. Thus, a mask has been taken into account, namely, voxels located closer from the crack than 3 voxels are considered “unreliable”, and on each projection all detector pixels whose ray crosses an “unreliable” voxel are disqualified from the quantitative evaluation of the residuals.

The Signal to Noise Ratio (SNR) of the projection over residuals  $\rho_0$  amounts to 36.1 dB when taking into account the flat field as well as possible. Equivalently, the standard deviation of the projection residuals amounts to 5.8% of the original projection dynamic range over the same area. Those numbers are obtained by assuming that there is an unknown displacement that allows to match them, and that the displacement is computed with the above algorithm. It also includes a 3-voxel wide mask positioned on the crack face to avoid crack projection artifacts. Figure 3 shows these residuals (without mask) over the region of interest. It is worth noting that when no motion is accounted for, the projection residual (which is a direct measurement of the quality of the reconstruction and of the inverse relationship between reconstruction and projection operators) computed over all 600 projections provides a RMS residual which is quite close to the above reported level (40 dB), although the absence of mask in the latter procedure forbids a direct comparison. The shape of the projection residual reveals that some spots appear on the bottom and top parts of the projections. They however have no temporal continuity, and hence they do not betray a problem from the reconstruction. They are initially present in the

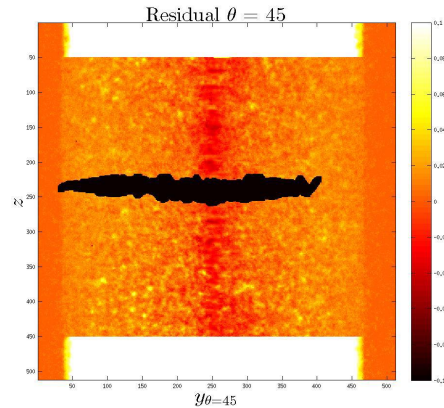


Figure 3. Projection residual in the reference state,  $\theta = \pi/4$ , accounting for a displacement that amounts to an RMS value of 0.21, 0.18, 0.16 voxel in the three directions

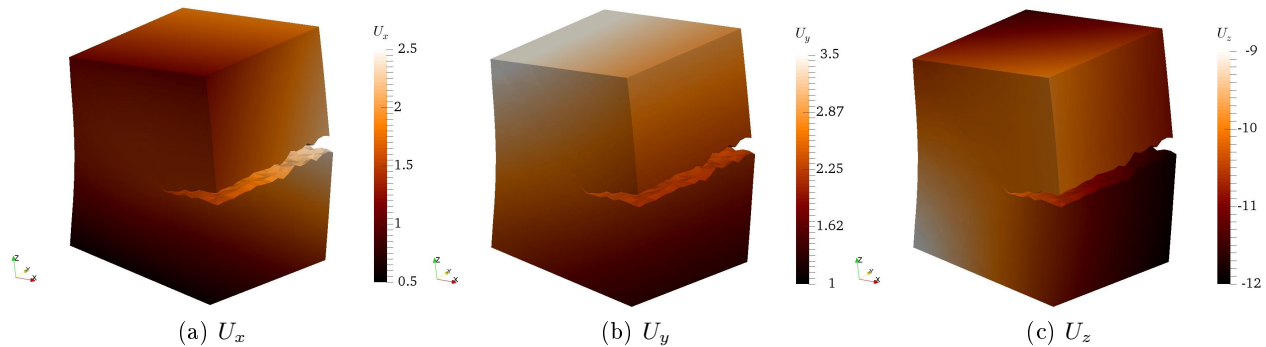


Figure 4. The three components of the displacement field along the tensile (longitudinal) axis shown on a deformed volume (displacements are amplified by a scale factor of 30)

sinogram, and hence it is an artifact of the radiograph acquisition, and not of the present treatment. Similarly, the vicinity of the rotation axis also shows a systematically lower residual. Hence the above mentioned displacement estimates provide a lower-bound to the uncertainty that can be expected (about 0.2 voxel) and that is not due to the procedure for displacement evaluation. It is also to be stressed that the magnitude of these artifacts is very small and appears negligible for most applications where a small fraction of a voxel cannot be noticed. It is only due to the fact that very small displacements are to be resolved that those artifacts become detectable and quantifiable.

## 5.2 Reference computed solution

After having estimated the quality of the registration in a (reference) case where the displacement is known to be 0, the actually deformed state is probed. As above, the twelve unknown boundary condition parameters are determined from only two orthogonal projections. The resulting displacement is shown in Figure 4.

The determined displacement field is also validated from the projection residual fields as shown in Figure 5. The signal to noise ratio amounts to 27 dB using the crack mask. This value is clearly lower than for the reference case (by about 10 dB). However, it should be noted that other sets of directions, for instance two projections at 0 and  $\pi/2$ , give an SNR of 31 dB and provide different scores to be compared to 37 dB when the projections are from the reference volume itself, *i.e.*, a decrease of 6 dB only. When considering the projection residuals in Figure 5, the similarity with the previous results is striking.

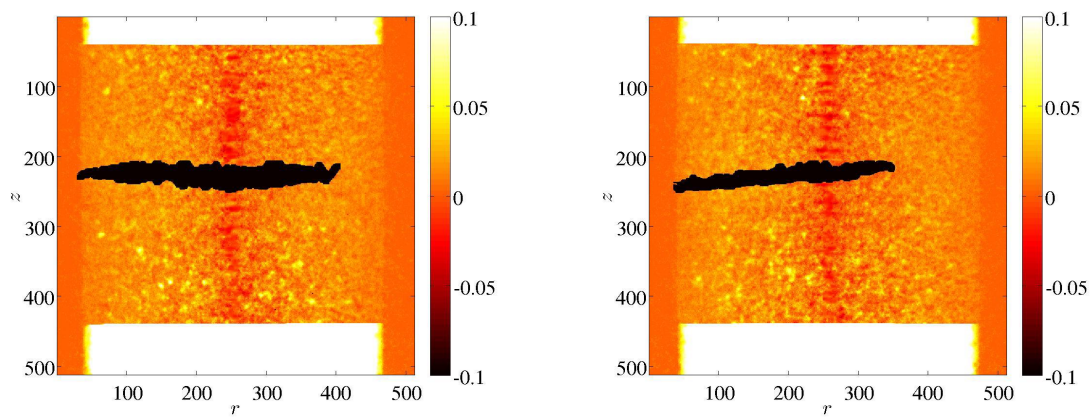


Figure 5. Projection residuals for (a)  $\theta = \pi/4$  and (b)  $\theta = -\pi/4$

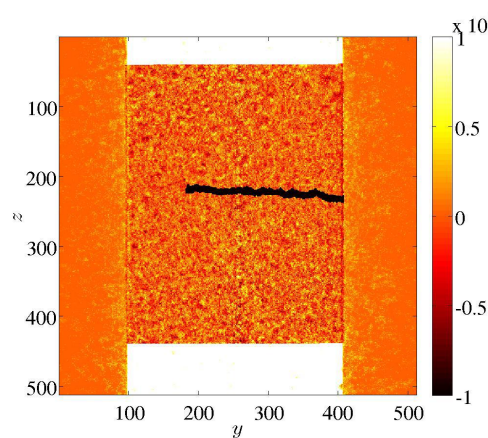


Figure 6. Difference between a slice  $x = 256$  through the reconstructed deformed volume and the equivalent slice in the reference volume advected by the measured displacement field

Because the test case was scanned using a classical procedure, 600 radiographs of the deformed volume are available. Hence, the deformed volume can be computed without any relationship with the reference volume. Thus the determination of the displacement is probed on the volume itself and not on the radiographs. Figure 6 shows the difference between a slice through the reconstructed deformed volume, and the same slice through the reference volume advected by the measured displacement field. Despite the use of only two projections, one can see that the motion is well (although not perfectly) captured. The SNR estimated on this slice is about 23 dB to be compared with 26 dB when the reference volume is used instead of the deformed one. One possible reason for an imperfect determination of the displacement is that the crack geometry that has been segmented from the reference volume may be too short (the crack trace vanishes close to the front), which results in too stiff a sample. A perspective to the present work is to allow for enriching the crack front position.

The high value of the SNR on the different residuals, be they estimated on the reconstructed volume or on the projections, shows that a good convergence has been achieved and that the P-DVC procedure can be deemed satisfactory.

### 5.3 Multiscale approach and rate of convergence

P-DVC (as DVC) remains an ill-posed problem. In particular if the displacement is too large as compared to an initial guess, convergence may fail. In order to capture large scale displacements, it has been shown

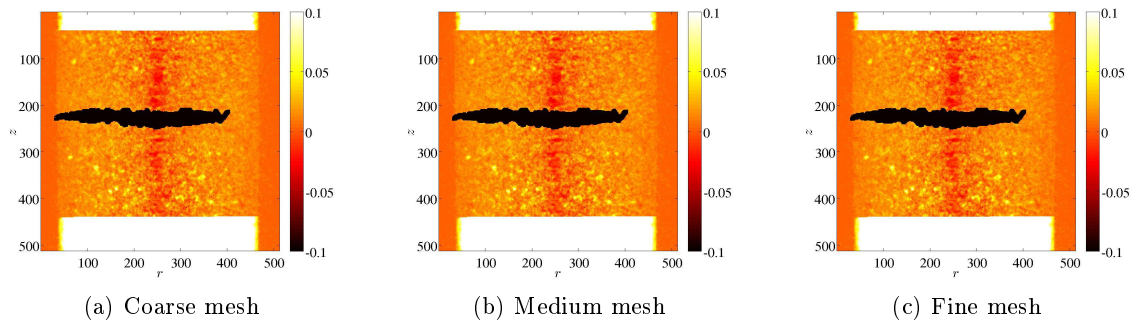


Figure 7. Projection residual estimated on the same projection angle  $\theta = \pi/4$  with the different meshes shown in Figure 2. (Note that (c) is identical to Figure 5(a), and reproduced here for an easier comparison)

that a multiscale procedure was very efficient.<sup>18</sup> The idea is simply to decrease the image definition (and the projections likewise) for instance by grouping  $2^3$  voxels into a supervoxel with the average gray level. In this case, the displacement expressed in supervoxels is smaller, and hence may be captured more easily (although less accurately). Moreover, the volume of data is very significantly reduced and hence the computation work load is small. This procedure can be applied recursively from a very coarse image, down to the original image, where the determined displacement field at a given level is used as an initialization for the next (finer scale) one.<sup>9</sup>

In the present case, such a multiscale procedure was not necessary, and convergence could be achieved without it. However, its use revealed beneficial in terms of convergence rate and overall computation cost.

#### 5.4 Sensitivity to reconstruction procedure

The very fact that the reference volume itself leads to significant residuals shows that the two operators  $\mathfrak{R}$  and  $\mathfrak{P}$  are only approximately inverse of each other. Noise in the radiographs contributes to this remaining residual but part of it is due to slight discrepancies in the description of voxels depending on the chosen algorithms. In particular, a popular and efficient way of reconstructing an image is the filtered back projection (FBP) algorithm that uses Fourier transform, whereas the projection step is generally performed in real space.<sup>10</sup> Algebraic reconstruction is an alternative choice to FBP. It consists in iteratively reducing the difference between the projection of the reconstructed volume  $\mathfrak{P}[f]$  and its measured projection  $P_f$ , very much in the spirit of the minimization of Equation (3) (although without displacement, and over the entire set of angles). Therefore, algebraic reconstruction and projection are closer inverses of each other as compared to FBP. The above obtained solution was processed using the algebraic reconstruction tool (SIRT algorithm) available in ASTRA.<sup>11</sup> One hundred iterations were performed after initialization from FBP reconstruction.

When an FBP algorithm is used, the projection residual SNR for the reference sample is significantly smaller, 22 dB, instead of 41 dB for  $\theta = \pi/4$ . Thus although it is difficult to evaluate the difference in reconstruction with bare eyes, the use of SIRT for reconstruction is observed to have a very significant impact for the present approach.

#### 5.5 Sensitivity to mesh fineness

The impact of mesh fineness is now addressed. In general, for DVC, as mesh fineness is increased, it is first favorable as long as it limits the approximation of picturing the actual displacement field by too coarse a mesh. It then becomes detrimental because of the progressive degradation of the conditioning of the system to invert and hence its increasing sensitivity to noise. However when an elastic regularization is used, the mesh fineness becomes irrelevant as soon as the solution to the elastic problem on the mesh is satisfactory.<sup>13,17</sup> In the present formulation, based on a diverging regularization length, only twelve degrees of freedom are considered and this number is independent of the mesh. Thus it is expected that the mesh fineness plays a very modest role.

This anticipation is checked against three meshes shown in Figure 2 by studying the projection residuals (see Figure 7). The SNR obtained for the coarse, medium and fine meshes are quite comparable, and the similarity

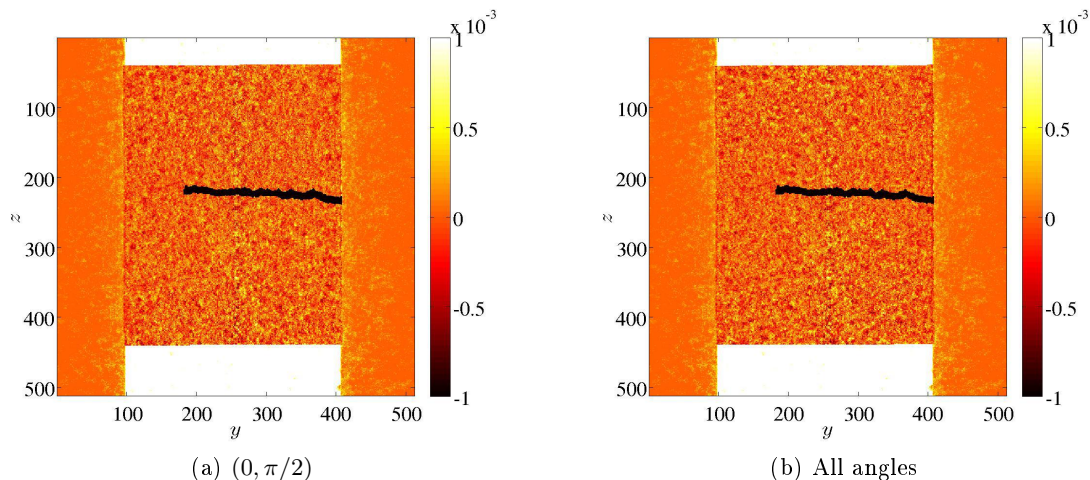


Figure 8. Difference between reconstructed deformed volume slice,  $x = 256$ , and the corresponding slice through the reference volume advected by the displacement field. (a) shows the residual computed from the displacement estimated from the  $(0, \pi/2)$  projection directions, while for (b) displacements were estimated with the entire set of 600 projections. These residuals are to be compared with Figure 6 obtained for  $(-\pi/4, \pi/4)$

between them is striking. This suggests that the acquisition artifacts (*i.e.*, bright spots in the projection residuals) are much more limiting than the displacement determination.

## 5.6 Sensitivity to number of projections and their angle

When two projections are chosen, a natural question to ask is the choice of angles. These orientations are to be chosen orthogonal to each other in order to maximize the independence of the two views. The sample cross-section is here rectangular, and if  $x$  and  $y$  are the eigen axis of the sample section, one natural choice is to choose orientations making an angle  $(-\pi/4, \pi/4)$  with respect to, say,  $x$ . The above result were obtained for such a choice. Alternatively, one may opt for  $(0, \pi/2)$ . It is difficult to find arguments for one choice or another. Phase contrast (an artifact occurring for large absorption contrast) is expected to be much more pronounced for  $(0, \pi/2)$ , and since such a phenomenon is not included in the projection/reconstruction procedures, one may wish to prefer the first set. A weakness of the latter though comes from the crack mask whose projection in the  $\pm\pi/4$  orientation obscures a larger area than the  $(0, \pi/2)$  choice. Alternatively, one may also have more than two projection directions. As an extreme choice, taking into account all 600 projections may be considered.

The residuals shown in Figure 8 are comparable in magnitude. The SNR obtained with the projections at  $(-\pi/4, \pi/4)$  is 27 dB whereas the choice of the two angles  $(0, \pi/2)$  gives 31 dB. When all angles are used, the SNR is 28 dB, thus even less than for the previous case using only two angles. Again, those numbers point to the conclusion that the limitation of the present approach is not the proposed methodology of measuring displacements from only two radiographs but rather acquisition artifacts.

## 5.7 Sensitivity to the mask

In the previous sections, a mask is used and applied on the reference volume with a width of 3 pixels around the detected crack surfaces. The reason for this mask is to prevent the effects such as phase contrast that are not included in the reconstruction (or projection). However it also masks potentially useful parts, especially in the vicinity of the crack. To check the influence of this mask on the procedure, different variants have been tested and compared, namely, no mask, a mask on the projection that has a fixed height of 36 pixels and is 20 pixel longer then the crack projection, and a mask applied on the reference volume and advected with the sample displacement field.

The black region seen in Figure 9 corresponds to the projection of voxels that are intersected by the crack surface. This region used to be larger in Figure 5a because a 3-voxel wide surrounding was masked on the

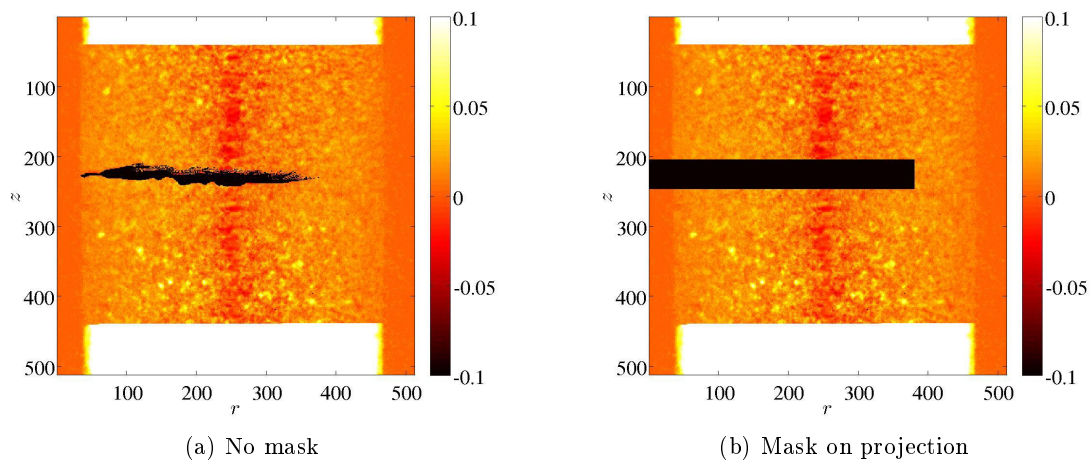


Figure 9. Projection residual without mask or with a fixed mask on the projection. These residuals are to be compared to Figure 5(a) where the mask is defined on the reference volume and advected with the displacement field

reference volume. For the three considered variants of masks, the same SNR of 27 dB is observed. Surprisingly, when the same mask sensitivity is probed on the reference volume, it is seen that with no mask the SNR drops from 36 dB to 27 dB, whereas the projection mask does not alter the SNR still equal to 27 dB.

## 6. CONCLUSION

The present paper introduces a new approach to track the time evolution of a 3D-imaged solid based on X-ray computed tomography. Once the reference solid has been fully imaged, the proposed approach is to capture its time evolution from its kinematics, which itself is to be read from a few projections. Moreover, in order to further reduce the number of parameters needed to describe its motion, a suited regularization technique is proposed, which uses a finite-element solution of an elastic homogeneous problem. The weight of regularization can be continuously tuned.

This procedure was tested on real tomographic data acquired on a synchrotron beamline for the analysis of a cracked specimen. It was chosen to give a very large weight to regularization so as to reduce the number of kinematic parameters to just a few (*i.e.*, 12), and hence the number of used projection to no more than two. The technique was successfully applied to analyze a deformed state under load. Many variants were tried in order to check the uncertainty of the approach. It turned out that the uncertainty is limited by artifacts of the data acquisition, much more than from the determination of the kinematics itself.

Such approach is to be further developed to investigate additional degrees of freedom such as those defining the crack shape geometry. For a single instant of time, the present approach cannot be reduced to less than two projections. However, for a continuous time evolution, using the temporal smoothness of the kinematics suggests to estimate it from a set of radiographs acquired at different instants. This opens the way to a software based investigation of “4D tomography.”

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